



جمهورية مصر العربية
وزارة التربية والتعليم والتعليم الفني
وحدة تشغيل وإدارة مدارس التكنولوجيا التطبيقية

Student guide

Physics

Year 3

(STEM)

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		TPC1.2	Understand electrically charged particles are released from material	
		TPC1.3	Study Einstein's relation	TPK2 Compton Scattering And X- Rays & LASER
		TPC1.4	Understand different between the characteristic radiation and the continuous radiation of X-rays	TPK3 classification Of Elements According To Electric Conductivity And Doping With Donors And Acceptors
		TPC1.5	Understand different between the spontaneous radiation and the stimulated radiation	TPK4 The P-N Junction (Diode) And Bipolar Transistor (P-N-P) And (N-P-N) And Logic Gates
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TP2	Explain the electric conductivity of semiconductors ,PN junction, Transistor and Logic gates	TPC2.1	Explain the difference between the two kinds of semiconductors (pure - impure, or intrinsic/extrinsic).	
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		TPC2.3	Compare with drawing between forward and backward connections of P-N junction. 2. Give examples of P-N junction applications	
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week	Unit	Third secondary Second term 2023
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Unit one

Modern physics

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Dear student, By the end of this lessons you should have the following skills and knowledges

Understand the difference between particle theory and wave theory

Explain the phenomena of black body radiation, photoelectric effect and Compton effect

Identify the structure of Cathode Ray Tube

Identify the properties of photon

Identify the properties of X- rays

Apply properties of X- rays

Understanding the characteristic and continuous spectrum of X- rays

Differentiae the spontaneous and stimulated emission

Identify the properties of laser

Lesson 1

Black body radiation phenomenon

Introduction

Radiation

Radiation is the transfer of heat by electromagnetic waves such as visible light, infrared, and ultraviolet radiation.

All objects radiate energy continuously in the form of electromagnetic waves produced by thermal vibrations of the molecules.

Wave particle duality

a) The wave particle duality means that

- 1) The wave behave as a particle
- 2) The particle behaves as wave

b) What is the difference between classical physics and modern physics?

Classical physics (macroscopic model)

It is that explains everything in our daily life and our common experiences.

Modern (quantum) physics (microscopic model)

It is that deals with scientific phenomena in the universe which might not be directly observed in our daily life, and classical physics cannot explain, especially

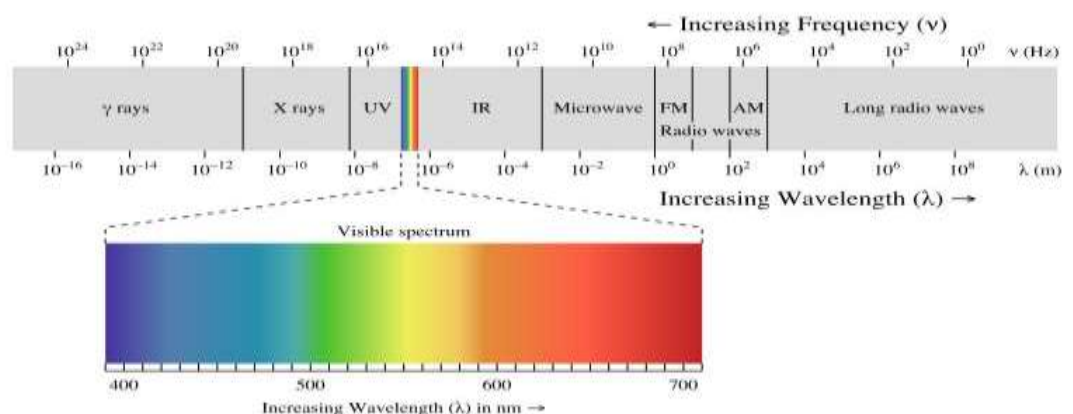
- 1) When we deal with atomic and subatomic systems.
- 2) Explains all phenomena involved in electronics which is the basis for all modern electronic and communication systems.
- 3) Explains chemical reactions on the level of the molecule.

(Some of such reactions were photographed by Ahmed Zewail using a high speed laser camera. Such work entitled him to earn the Noble Prize In chemistry in 1999).

Electromagnetic spectrum

The common features

- 1) They are vibrating **electric and magnetic fields** that moves through **space** or any other **medium**.
- 2) They are **transverse waves** only.
- 3) They **don't need medium** to travel through it.
- 4) They **propagate in space** (as in air) **at the same speed** (c) = 3×10^8 m/s **while varying in frequency (and wavelength)** and propagate in material media with a smaller velocity depending on the medium.



The Visible light of this spectrum

- * It is the part that can be seen with the human eye
- * It is only a very small part of this spectrum. ($\lambda=400\text{nm}-700\text{nm}$)

Black body radiation**Introduction:**

- 1) Each surface absorbs and reflects different frequencies of light. **White surfaces** reflect all colours. **Black surfaces** absorb all colours.
- 2) When any radiation is absorbed its energy is changed into heat energy. This means that black objects that absorb radiation easily heat up quickly.
- 3) An object at any temperature emits electromagnetic radiation, called **thermal radiation or (blackbody radiation)**

Hot bodies emit light and heat such as the Sun (**yellow**), a burning coal (**red**)

Why do different objects look different colours?

Because the dominant color of light (the wavelength) emitted from these sources varies according to **the temperature and properties of the object.**

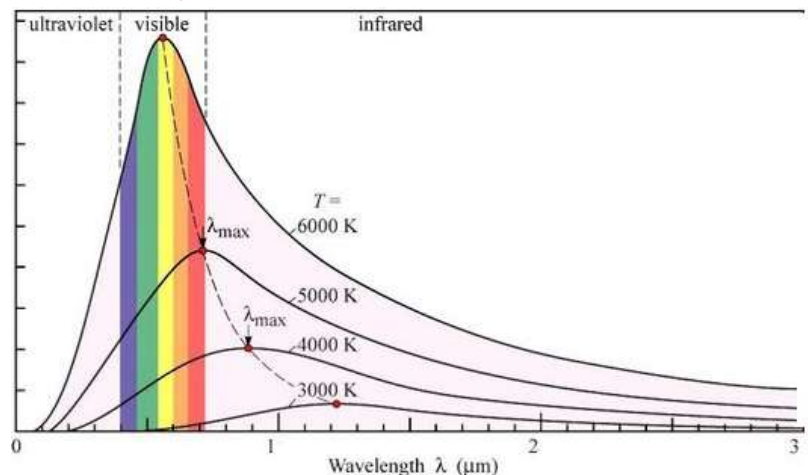
- 1) Electromagnetic source does not emit all wavelengths equally, but the intensity of radiation varies with wavelength (or the frequency).
- 2) The color changes according to the temperature of the source.
- 3) At low temperatures, the wavelengths of the thermal radiation are mainly in the infrared region and hence not observable by the eye. As the temperature of an object increases, the object begins to glow red. At sufficiently high temperatures, it appears to be white, as in the glow of the hot tungsten filament of a light bulb

Planck's Distribution (Planck's curve) :

The distribution of the radiation intensity with wavelength is called Planck's distribution

From the Planck's distribution

- 1) The wavelength tends to infinity (very larger) or to zero (very small) the intensity of radiation tends to zero.



	The sun radiation (T = 6000 K)	The lamp radiation (T = 3000 K)
temperature	(T = 6000 K)	(T = 3000 K)
The wavelength at the peak	$\lambda_{\max} = 5000 \text{ \AA} \text{ (500 nm) (0.5 μm)}$ (within the visible region)	$\lambda_{\max} = 10000 \text{ \AA} \text{ (1000 nm) (1 μm)}$ (within the infra-red region)
the emitted energy	40% is a visible light 50% is a heat radiation 10% is a different region	20% visible light 80% heat radiation

- 2) The previous curve (Planck's distribution) is repeated for all hot bodies which emit continuous radiation not only the sun but also the earth and all bodies even living creatures. But the earth being a **non glowing body** it absorbs the radiation from the sun

- 3) The intensity of radiation is related to the wavelength

$$E = h\nu$$

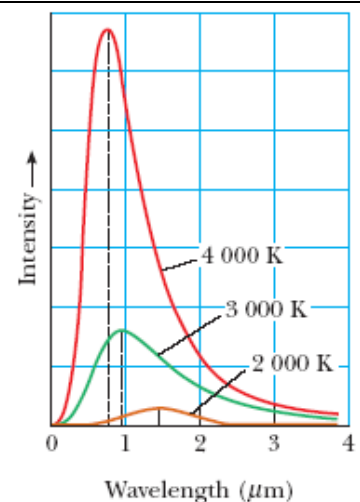
$$\nu = \frac{c}{\lambda}$$

$$E = h \frac{c}{\lambda}$$

- 4) The maximum intensity of radiation increases by increasing the temperature

- 5) we can note the previous relations in our daily life

- a) **when a piece of iron is heated to become self-luminous, the apparent color of the emitted radiation shifts initially from red to orange to yellow to blue and finally to white as the temperature increases.**
- b) **The sun is yellow because its surface temperature is about 6000 K. Some stars glow blue-white, because their surface temperature can be as high as 20000K.)**



Wien's Law

Wien's law describes one of the **relations between the emission spectrum of a black body and its temperature**. it states that

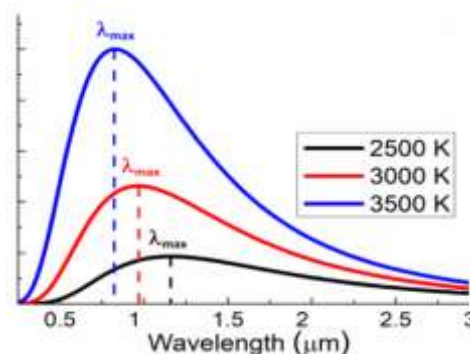
(The wave length at maximum radiation intensity ($\lambda_{\max}$) is inversely proportional to the temperature)

OR

(The wavelength ($\lambda_{\max}$) at which the peak of the curve occurs is inversely proportional to temperature).

$$\lambda_{\max} \propto \frac{1}{T}$$

$$(\lambda_{\max} T = \text{constant})$$



Therefore, the higher temperature, the smaller the wave length of the peak

Exercise

A body at temperature 1500 K radiates out maximum energy at the wavelength 20000 Å. If the sun radiates out maximum energy at 5000 Å, calculate the temperature of the sun.

Applications based on infra-red emitted from bodies (Thermal imaging) (remote Sensing):

- 1) Satellites and airborne which map and photograph the earth's surface. And the scientists analyze these images to determine the natural earth resources
- 2) Military purposes such as night vision system, which detect and image moving objects in the dark due to the heat radiation which these objects re - emit
- 3) In medicine (tomography – embryology)
- 4) In criminology (since the heat radiated from a person still for a while even after the person has left)

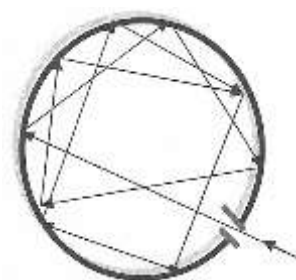
Black body radiation

Definition:

It is an ideal system that absorbs all radiation (100% of the radiation that hits it) falling on it. Then re-emit this radiation totally again

Description:

Enclosed cavity with a small hole and the inside of the cavity appears black **because all of the radiation within the cavity remains trapped due to multiple reflections. Only a small part of it leaks out, which is called blackbody radiation**



Note that:

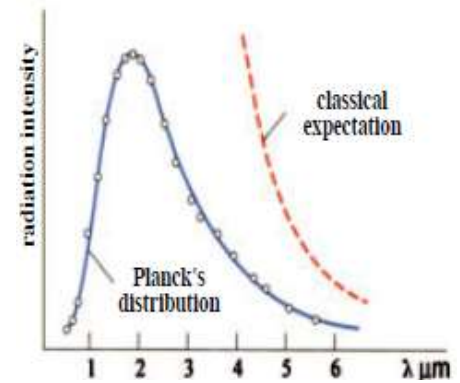
- 1) Black body is a Perfect absorber and perfect emitter
- 2) Black body reflects no radiation and appears perfectly black.

The explanation of Planck's distribution

$$E = h\nu$$

$$\nu = \frac{c}{\lambda}$$

$$E = h \frac{c}{\lambda}$$

**First: Classical explanation (The wave theory of light)**

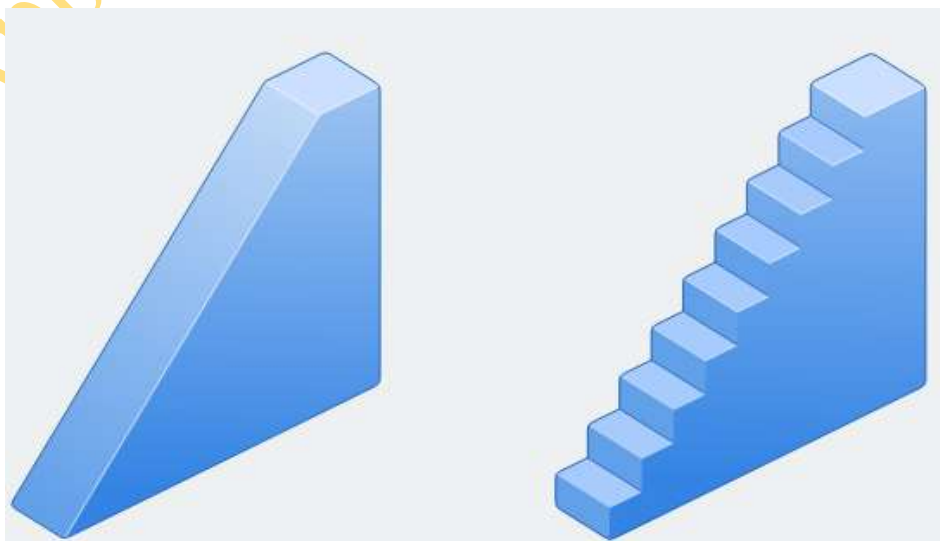
- 1) The intensity of radiation increases with the increasing of frequency. This intensity became infinite at high frequencies.
- 2) The classical physics cannot explain that why at high frequency the radiation intensity decreases.

Conclusion:

- a) At long wavelengths (low frequency), classical theory is in good agreement with the experimental data.
- b) At short wavelengths (high frequency), however, major disagreement exists between classical theory and experiment.

Second: Planck's explanation (The particle theory of light)

In 1900, the German physicist Max Planck (1858–1947) explained the Planck's curve by proposing that the radiation was made up of small units (or packets) of energy, each he called quantum (or photon) and the energy of electromagnetic waves is *quantized* (**$E = n h \nu$ where $n = 0, 1, 2, \dots$ is an integer**) rather than continuous. This means that for each temperature, there is a maximum intensity of radiation that is emitted in a blackbody object, corresponding to the peaks, so the intensity does not follow a smooth curve as the temperature increases, as predicted by classical physics. Thus energy could be gained or lost only in integral multiples of some smallest unit of energy, a quantum (the smallest possible unit of energy). Energy can be gained or lost only in integral multiples of a quantum. The atom does not radiate as long as it remains in one energy level. But if it shifts from a high energy level to a lower energy level, it emits a photon whose energy $E = h\nu$. (Photons of high frequency have high energy and vice versa)



Lesson 2

Thermionic emission

Introduction

There are many ways to eject (escape) electrons out of a material.

1) **Thermionic emission** (emission of electrons from a heated metal's surface (by heating))
Cathode Ray tube (C.R.T) that used in TV and computer monitors is an application based on **Thermionic emission**

2) **The photoelectric effect** (emission of electrons from a metallic surface when the light energy falling on it)

Photoelectric cell is an application based on **the photoelectric effect**

Thermionic emission

A metal contains positive ions and free electrons which can move around inside the metal but cannot leave it, due to the attractive forces of the surface

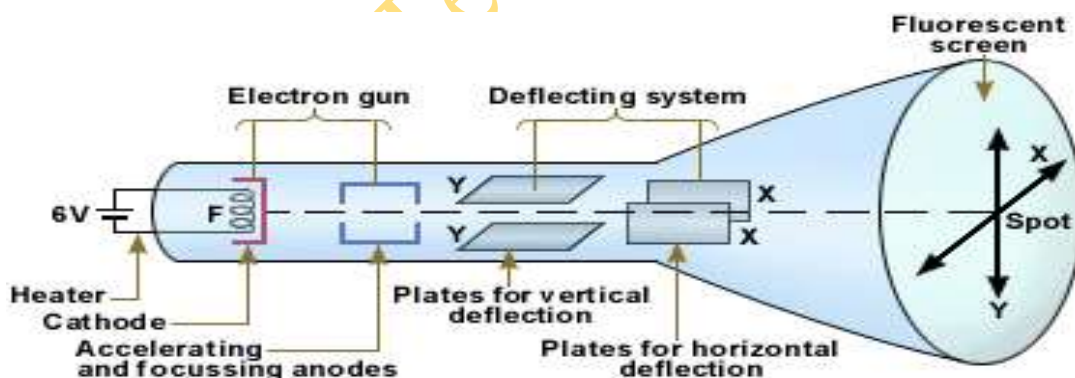
Surface potential barrier

(It is the attractive force between the positive ions and free electrons in metal which make the electrons cannot leave it)

But some of these electrons can escape if given enough energy in the form of heat or light.

This is the idea behind the cathode ray tube (CRT), which is used in TV and computer monitors

The cathode ray tube (C.R.T)



Scientific idea:

It depends on the **Thermionic emission**

(Emission of electrons from a heated metal's surface (by heating))

Uses

It is used in TV and computer monitors.

The operation of cathode ray tube

The tube consists of a metal surface called the cathode (negative electrode) which is heated by a filament. The electrons are produced due to heat so called (electron gun) (E-gun). Some electrons may overcome of attraction at the surface (potential barrier surface). These electrons liberated from metal and picked up by the screen which connected to anode (positive electrode).

When electrons hit the screen, they emit light which varies in intensity from one to another on the fluorescent screen depending on the intensity of the electrical signal transmitted. The signal controls the intensity of electron beam emitted from the E – gun through a negative grid in its way. The E–gun can be controlled by electric and magnetic field to sweep the screen point by point (raster) until the frame is completed.

Functions of CRT parts

1) Filament

It is used to heat up the cathode

2) Grid

Control the number of electrons in the electron beam

3) Anode

It is used to accelerate the electron beam towards the screen

i.e. it Picks up the electrons which are liberated from the cathode, thus causing current in the external circuit

4) Electron gun

It is an instrument to produce a beam of electrons at high speed

5) Fluorescent Screen

Emits green light when the electron strike it

Notes

- 1) The cathode is heated by an electric current passes through the filament.
- 2) The intensity of this illumination (illumination of the fluorescent screen) depends on the kinetic energy of the incident electron and the kinetic energy of the electron depends on the potential difference between the cathode and the anode where

$$(K.E. = eV = \frac{1}{2} m_e v^2)$$

Where

e : the electron charge (1.6×10^{-19} C)

V : potential difference between anode and cathode

m_e : the mass of the electron.

v² : the square of electron velocity.

Lesson 3

Photoelectric effect

Introduction

There are many ways to eject (escape) electrons out of a material.

3) **Thermionic emission** (emission of electrons from a heated metal's surface (by heating))
Cathode Ray tube (C.R.T) that used in TV and computer monitors is an application based on **Thermionic emission**

4) **The photoelectric effect** (emission of electrons from a metallic surface when the light energy falling on it)

Photoelectric cell is an application based on **the photoelectric effect**

Photoelectric effect:

(It is the emission of electrons from a metallic surface when the light energy falling on it)

i.e. the Light energy converted to electrical energy and the emitted electrons are called **photoelectrons**.

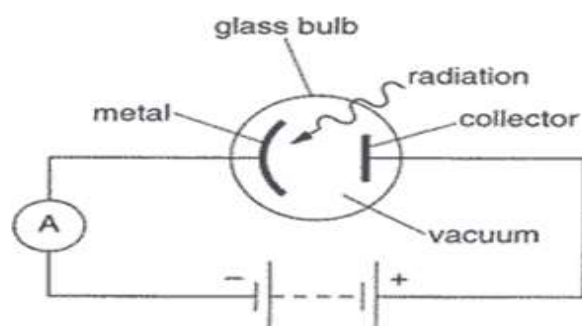
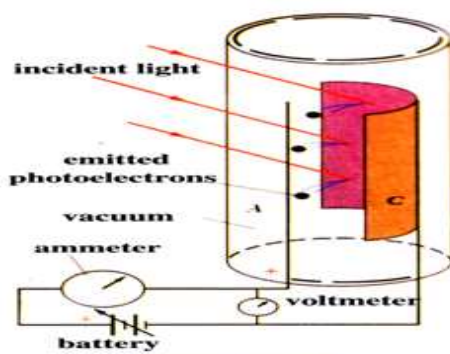
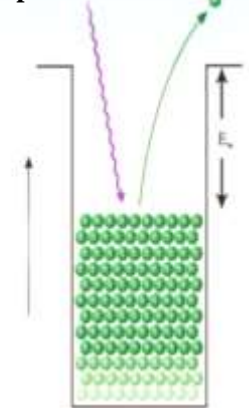
Photoelectrons

An electron released or ejected from a substance by photoelectric effect

Experiment

When light falls on a cold cathode, current flows in the circuit .this means electrons are emitted due to light (photoelectric effect)

Incident photon



The explanation of the photo electric effect

Einstein's theory (The correct explanation for photoelectric effect)

Einstein used Planck's idea of quantization of electro-magnetic radiation to explain the results of experiments on the photo-electric effect

$$K. E_{max} = h\nu - E_w$$

$K. E_{max}$: The kinetic energy of the emitted electrons

$h\nu = E$: the photon energy

h : Planck's constant = 6.625×10^{-34} J.s

ν : photon frequency

E_w : the work function of the metal

The work function of the metal

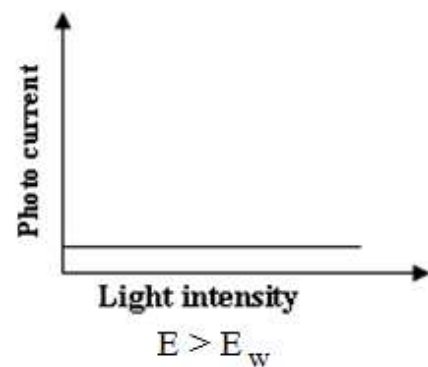
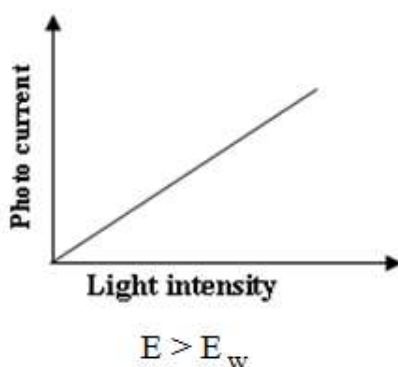
(It is the minimum energy needed to free (release) the electrons from the surface of the metal)

To free (emit) an electron with kinetic energy from the metal, the energy of the incident photon must be greater than the work function

The kinetic energy of the emitted electrons is independent of the intensity of the illuminating source of light but it depends upon the frequency of the incident light and the work function of the metal where it increases by increasing the frequency of the light.

The relationship between light intensity and photocurrent

When the photon energy (E) is **greater than** the work function (E_w), the number of photoelectrons emitted per second is directly proportional to the intensity of the radiation. The photocurrent increases with the intensity of the incident light but when the photon energy is **less than** the work function, there is no electrons will be emitted (the photocurrent will be zero)



Excercise

- 1) A metal has a work function 8×10^{-19} J. Calculate the maximum kinetic energy of electrons emitted from the metal when light of frequency 2×10^{15} Hz is shone on the surface (Given that $h = 6.625 \times 10^{-34}$ J.s, $C = 3 \times 10^8$ m/s)
- 2) When monochlight of wave length 663 nm is incident on a metal plate whose work function 3.52×10^{-19} eV will elerons be emitted or not ?
(Given that $h = 6.625 \times 10^{-34}$ J.s, $C = 3 \times 10^8$ m/s)

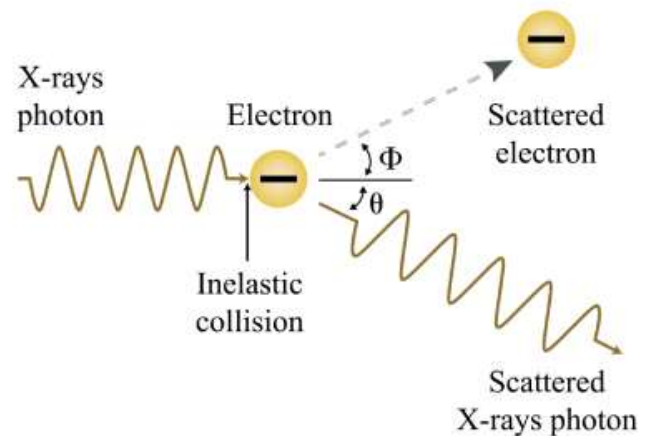
Applied Technology Schools

Lesson 4

Compton Effect

Compton Effect

When a photon of high frequency (x- rays or gamma rays) collides with a free electron, the photon frequency decreased and changed its direction; also the electrons velocity increased and changed its direction.



Compton Proved that

The particle nature of light (we must consider a photon like the electron as a particle with linear momentum (it has mass and velocity))

How Compton Prove the particle nature of light?

He noticed that from the experiment, **the scattered light has a wavelength greater than that of the incident light.** (This observation could not be explained by the classical theory of light, where the wavelength does not change).

How Compton explained this problem?

By considering light as a photon with energy given by Planck's **assumption**

The electromagnetic radiation consists of photons which can collide with electrons

In this collision Compton was able to verify

1) The conservation of momentum

The total linear momentum before collision = the total linear momentum after collision

(The total linear momentum is the same before and after collision)

2) The conservation of energy

The sum of energy of photon and electron before collision = the sum of energy of photon and electron after collision.

(The total energy is the same before and after collision)

Note that

1) The wavelength of the scattered photon ($\lambda_{\text{scattered}}$) is greater than the wavelength of the incident photon ($\lambda_{\text{incident}}$)

i.e. (the scattered (reflected) light has lower frequency and longer wavelength because it lost energy and momentum to the electron)

2) The velocity of the electron increases after collision and also kinetic energy

$$(K.E. = \frac{1}{2}mv^2)$$

3) The photon speed is the same before and after collision

In finally

Black body radiation, photoelectric effect and Compton effect are conclusive evidence that light behaves as particles (the particle theory of light) (light consists of photons)

The basic properties of photons

- **Photons: Electromagnetic radiation travels in space in the form discrete packets of energy (quanta)**
- **Photons travel in straight line with the speed of light ($c = 3 \times 10^8$ m/sec in vacuum)**
- **Energy of photon is given by**

$$E = h\nu = \frac{h c}{\lambda}$$

- **Momentum of photon is given by**

$$P_L = \frac{E}{c} = \frac{h}{\lambda}$$

- **Photons are electrically neutral particle and they are not deflected by the either electric or magnetic field**

Lesson 5

X- Rays

Introduction:

What are x- rays?

They are invisible electromagnetic waves of short wavelength ($10^{-3} \text{ \AA} - 100 \text{ \AA}$), located between gamma rays and UV rays. They were discovered by Roentgen in 1895.

What are X-rays?

- 1) They are a part of the electromagnetic spectrum
- 2) They have a very high frequency and also a very high penetrating power
i.e. they are characterized by frequencies higher than those of ultraviolet radiation and having the ability to penetrate most materials with relative ease.
- 3) They have very short wavelengths of the range 10^{-13} to 10^{-8} m.
- 4) They are lie between ultraviolet (UV) and gamma rays (γ -rays)
- 5) They were first discovered by **Rontgen**, he called X-rays because their nature was at first unknown

Properties of X-rays

- 1) They can penetrate media easily
- 2) They can ionize gases
- 3) They diffract in crystals
- 4) They affect sensitive photographic plates

Coolidge tube

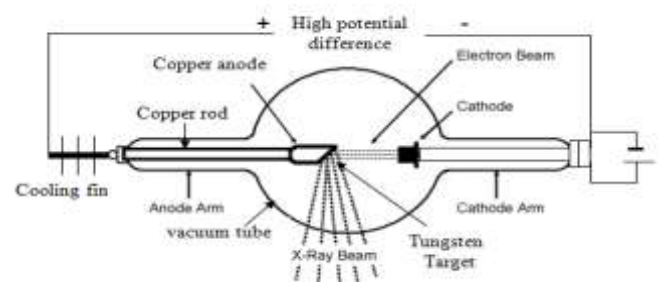
Use: Production of X-rays

Structure

- 1) **Filament** (cathode) (source of electrons)

The filament is generally made out of tungsten and is coiled to provide high resistance to the passing current.

- 2) **Solid copper rod** (anode)
- 3) **Heavy element** (molybdenum) (target element) (anode)
- 3) **An external high DC voltage source** (A high potential difference between the cathode and anode)
- 4) **A battery** that supplies high current heats the cathode filament
- 5) **Vacuum tube** to avoid the collision with air molecules
- 6) **Cooling fins** (to reduce heating effect)



Theory of action

How to produce X-rays by using Coolidge tube?

The electrons emitted from the filament, experience the potential difference and are accelerated towards the target. Thus the electrons gain energy, when an electron collides with the target element. Part-**if not all**-of its kinetic energy converted to X-rays

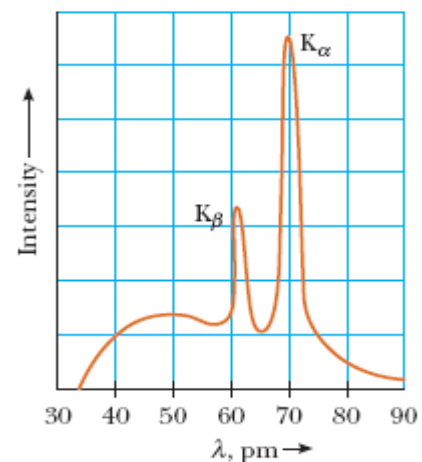
Spectrum of X-rays

Analyzing a beam of X- rays generated from target to components of different wavelengths, **we find that the spectrum consists of two parts:**

- The continuous spectrum** of all wavelengths (within a certain range) regardless the target material
- The line spectrum corresponding to specific wavelength** characteristic of the target material called the characteristic x – rays radiation

1) Characteristic radiation

- The line spectrum is generated when an electron collides with an electron close to the nucleus of the target material atom.
- The electron in an atom of the target metal in a low energy level is excited to a higher energy level because it gains sufficient energy and is replaced by another electron from a higher level to fill the space in the lower energy level.
- The difference in energy appears as radiation with a definite wavelength



Note that:

- The wavelength of the characteristic radiation** (at the peak) does not depend on the voltage difference but **depends on the kind of target element**
- The higher the atomic number of the target element, the shorter the wavelength of the characteristic radiation
- At low voltage differences, no characteristic radiation is produced.

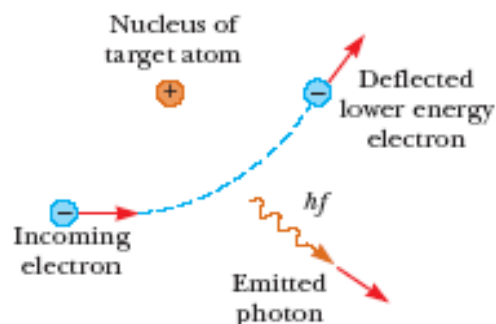
Appl

2) Continuous radiation

a) This radiation is generated when the velocity of the colliding electrons is reduced upon passing through a retarding electric field due to the electrons of the target material.

b) This energy decrease is due to repulsion, collision and scattering. Hence, an electromagnetic radiation is generated based on Maxwell - Hertz theory.

i.e. when the colliding electron passes close to a nucleus it is deflected, the change of direction means the electron has been accelerated. An accelerating charged particle emits electro-magnetic radiation



c) This radiation is continuous and called **soft X-rays**

d) The continuous radiation is sometimes called bremsstrahlung, a German word meaning “braking radiation,” because electrons emit radiation when they undergo acceleration inside the target.

Note that

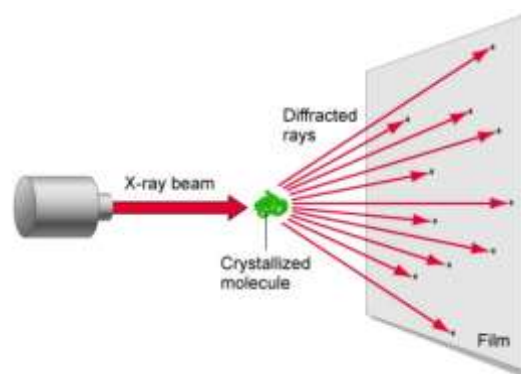
- 1) **The minimum wavelength** (maximum frequency) of the continuous radiation **depends on the voltage difference between the cathode and the anode (the accelerating voltage)**
- 2) Soft X-rays are produced when high-speed electrons are suddenly decelerated. When electrons have been accelerated through a voltage V , the shortest-wavelength radiation that can be produced is given by the following relation

$$eV = \frac{hc}{\lambda_{\min}} = h\nu_{\max}$$

Important applications of X – rays:

1) Studying the crystalline structure

X-rays have **ability to diffract**, as they penetrate materials, this is why x-rays are used in studying the crystalline structure of materials. The atoms in the crystal act as a diffraction grating (which is a generalization of diffraction from a double slit). Bright and dark fringes form, depending on the difference in the optical path.



2) Detecting the defects in metallic structure

X-rays have **a great penetrating power** this is why x-rays are used to detect defect in metallic structure

3) In the medicine

They are used in imaging bones and fractures and some medical diagnosis



Lesson 6

Laser

Introduction:

LASER is the **acronym** for **L**ight **A**mplification by **S**timulated **E**mission of **R**adiation

Spontaneous Emission and Stimulated Emission

Introduction

1) The atom has many energy levels

a) The lowest energy state ($n=1$) is called the **ground state** (E_1) of the atom (The lower the energy, the more stable the atom)

b) The **higher** energy states ($n=2, 3 \dots$ etc) is called the **excited states** (E_2)

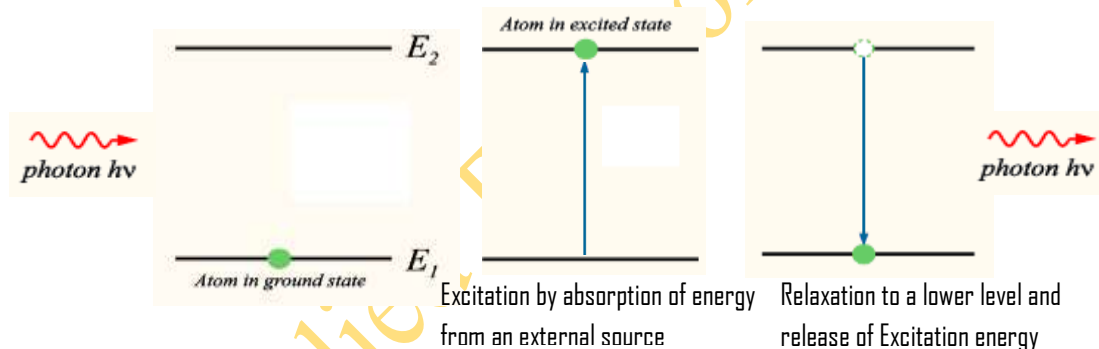
i.e. When an electron is in a higher energy orbit (i.e. $n=2$ or higher) the atom is said to be in an excited state

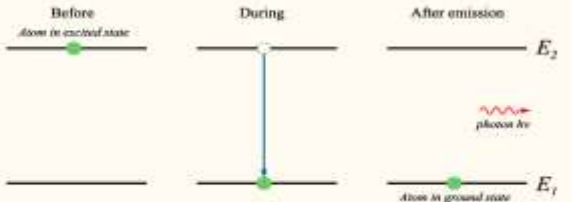
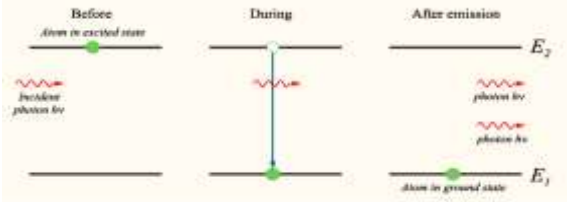
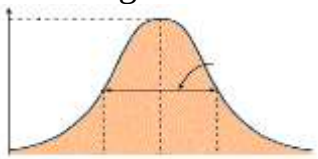
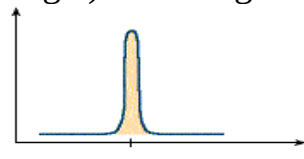
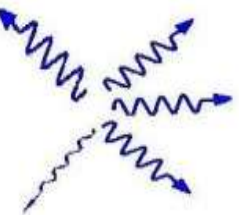
2) $E_2 - E_1 > E_3 - E_2$

3) According to assumptions of quantum mechanics relaxation and excitation process are simultaneously

If we shine a photon with energy $h\nu = E_2 - E_1$ on the atom in ground state (E_1), the atom absorbs this photon and gets excited to higher state (excited state) (E_2). The excited atom returns to the ground state and gets rid of this excitation energy in the form of an emitted photon.

Light	LASER
Amplification	
Stimulated	
Emission Radiation	

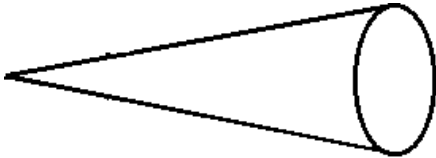
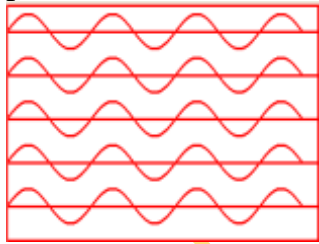
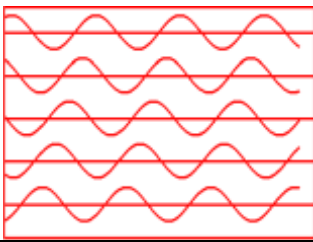


Spontaneous Emission	Stimulated Emission
Occurs when the atom relaxes from an excited state to a lower state after lifetime interval is over (nearly 10^{-8} s) emitting spontaneously the energy difference in the form of a photon without the effect of an external photon.	Occurs where an external photon of energy $E_2 - E_1$ passes by excited atom at E_2 to relax it to E_1 before life time is over
	
The emitted photons have a wide range of wavelengths 	The emitted photons are monochromatic (single) wavelength. 
The emitted photons propagate randomly 	The emitted photons are coherent and propagate in one direction as a collimated parallel beam. 
The intensity of photons decreases according to the inverse square law. This is called spreading. While collisions with particles is called scattering. In ordinary light sources both spreading and scattering occur.	The intensity remains constant over long distances contrary to the inverse square law. It has been possible to send a laser beam to the Moon and receive it back, without much loss, despite the long distance involved. Spreading effect is nil and limited scattering takes place
This is the dominant radiation in ordinary light sources.	This is the dominant radiation in laser sources.

Properties of a laser beam (Characteristics of Laser beam)

1) Monochromaticity	The radiation is almost one wavelength, as can be measured by the very narrow spectral width)
2) collimation	a) Very small divergence of the beam. b) The beam is almost a parallel beam and move in one direction in space (Directionality)
3) coherence	Photons are emitted at the same time, in phase and in the same direction, to form sharp laser beam
4) intensity	a) The laser is not subject to the inverse square law, b) the intensity of radiation falling on unit area is constant, due to unspreading of its rays

The light emitted by lasers is different from that produced by more common light sources (ordinary sources) such as incandescent bulbs and fluorescent lamps.

Point of comparison	Laser beam	Ordinary light
Monochromaticity	the laser emits energy over a very narrow wavelength range, literally, one color	Each line in the visible spectrum in ordinary light sources includes a band of wavelengths
collimation	The diameter of the emitted light beam stays constant for long distance without much unscattering. Thus, energy is transmitted without much loss. 	The diameter of the emitted light beam increases with distance. 
coherence	Photons emanate coherently both in time and place, since they come out together at the same time sequence, i.e. the emitted Photons are in phase and in the same direction, 	Photons propagate randomly or incoherently. They emanate at different instants of time, and have inconsistent and varying phase. 
intensity	The laser does not obey the inverse square law, since the intensity of radiation falling on unit area is constant, the further away from the light source, due to unspreading of its rays.	This light obey the inverse square law, since the intensity of radiation falling on unit area decreases, the further away from the light source, due to spreading.

Note that

The ordinary color appears to have different shades to the naked eye because the ordinary light includes a bandwidth of wavelengths

Theory of the laser Action

Laser action depends on driving the atoms or molecules of the active medium into a state of Population inversion.

The two conditions necessary for laser action are

- 1) Population inversion
- 2) Stimulated emission

Firstly: the stimulated emission has the same frequency and phase as the incident radiation.

Secondly: A population inversion exists whenever more atoms are in an excited atomic state than in some lower energy state.

i.e. The number of atoms in the excited state exceeds the number of atoms in the lower state (i.e., more laser-active atoms or ions are in the higher level).

Amplification means that the Photons are increased in number going back and forth in the active medium, due to multiple reflections between two enclosing mirrors.

Main components of a laser

- 1) Active medium
- 2) Source of energy
- 3) Resonant cavity

1) active medium

(The effective material required to produce laser)

Collections of atoms, molecules or ions in the form of solid or liquid or gas

2) source of energy

(it is the responsible for exciting the active medium)

- a) Excitation by electric energy
- b) Excitation by optical energy (optical pumping)
- c) Thermal excitation
- d) Excitation by chemical energy

Note that:

Pumping process

Exciting some of active medium particles into high-energy state by using an external energy source

Pumping of the active medium is usually performed in one of the following forms:

- 1) Optical pumping (flash lamp)
- 2) Electrical pumping (electric gas discharge)(High DC voltage)
- 3) Chemical pumping (exothermic chemical reactions)

3) resonant cavity

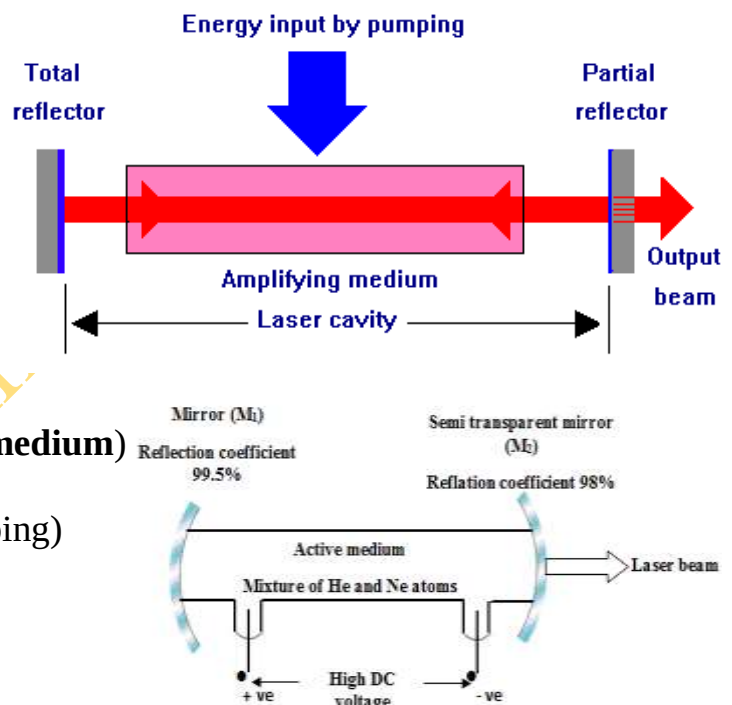
it is the container and the activating catalyst for amplification

1) External resonant cavity

Two parallel mirrors enclosing the active medium permitting multiple reflections leading to amplification

2) Internal resonant cavity

The ends of the active material are polished so as to act as mirrors



Note that:

- 1) The two mirrors act as **reflectors**, for light amplification
- 2) One of the two mirrors is semitransparent to allow some of the laser radiation to leak out
To stimulated emission takes place, it must be **Meta – stable excited level existed**

Meta – stable excited level

It is an excited energy level characterized by its lifetime is relatively long nearly 10^{-3} s

Laser Applications**Notes:**

- 1) There are different types and sizes of lasers. Laser light covers different regions of the electromagnetic spectrum from visible to UV and IR.
- 2) Some laser systems can focus a laser beam in a small spot, where energy might get so high as to melt- and even evaporate iron, or pierce diamond
- 3) There are lasers which may have enough energy to destroy missiles and planes in what is termed Star War.
- 4) Some applications of lasers also include holography and medical applications.

Laser has many applications in different fields such as:**Medical applications (highly collimated laser beam)****1) Treatment of retinal detachment (Welding and Cutting)**

- * The retina contains light sensitive cells; part of the retinal loses its function.
- * Unless quickly treated the eye may lose sight completely
- * Lasers are used for reconnecting the detached part with the layer underneath (in early stages)
- * The operation takes less time and effort than before. By using the thermal heat from the laser that cauterizes the points of detachment (endothermic)

2) Treatment of far and near sightedness cases**3) Other applications of lasers in medicine**

- a) Endoscopy where lasers with optical fibers are used for diagnosis and operative surgery
- b) General Surgery.

Industrial applications (highly collimated laser beam)

- 1) Industry, particularly fine industries
- 2) melt- and even evaporate iron, or pierce diamond

Military applications (highly collimated laser beam and coherence)

- 1) Precision guidance.
- 2) Smart bombs
- 3) Laser radar (LADAR).

Scientific research applications

- 1) Space research
- 2) Surveying (determining dimensions and areas)

Daily applications (Monochromaticity and coherence)**1) Communications**

Where optical fibers carry information - loaded laser beam instead of a wire carrying electrical signals

2) CD recording**3) Laser printing**

Laser beam is used to carry information from the computer to a drum coated by a photosensitive material. A toner is used to print off from the drum onto paper

4) Arts and laser shows and holography

Unit two

Modern electronics

Lesson	Page
Lesson 1: Semiconductors	26
Lesson 2 : Doping process	28
Lesson 3: The P-N junction (diode)	29
Lesson 4: Bipolar Transistor (P-N-P) and (N-P-N)	32
Lesson 5 : Logic gates	35

Dear student, by the end of this unit you should have the following skills and knowledges

Identify how to increase the electric conductivity of semiconductors

Explain pn junction (diode)

Explain transistor

Differentiate between the digital electronics and analogue electronics

Identify the five common logic gates

Use the transistor as an amplifier

Calculate the transport factor and current gain of a transistor

Lesson 1

Semiconductors

Introduction:

Today, Electronics and communication are now indispensable in our daily life.

- 1) TV, cellular (mobile) phone, computer, satellites and other systems are evidences for the vast progress in the applications of electronics and communications
- 2) They have become also an essential ingredient in modern warfare. Weapons do not fare from the point of view of fire power only, but guidance, surveillance, monitoring jamming and deception, called electronic counter measures (or ECM) play an important role in combat.
- 3) Also, in medicine whether in diagnosis, prognosis, or operations, electronics plays a key role.

In short there is no single field in all walks of life where electronics has no part, starting from e - games to e - warfare

Semiconductors

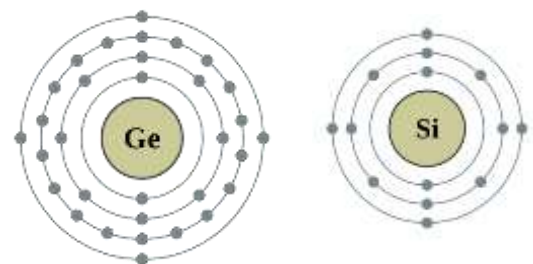
They are materials with conductivities somewhere between conductors and insulators.
(Silicon and germanium)

Semiconductors are divided into two categories:

- 1) Intrinsic Semiconductors (**Pure semiconductors**)
- 2) Extrinsic Semiconductors

Pure semiconductors

- a) Pure semiconductors have **low conductivity**.
- b) Pure semiconductors behave as **perfect insulator at 0K**, whereas as **temperature increases their conductivity increases**
- c) Pure **silicon (Si)** and **germanium (Ge)** crystals in their natural states are **pure semiconductors**
- d) Silicon and germanium have a unique property in their electron structure each has **four electrons in its outer orbit**



Note

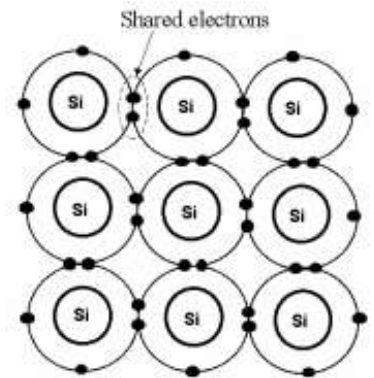
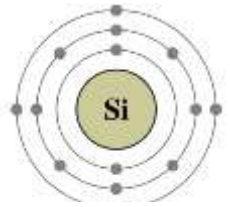
- 1) Silicon is one of the important and common elements in the universe
- 2) Crystal of pure Silicon consists of Silicon atoms bound together in covalent bond

The crystal

(It is a regular arrangement of atoms in the solid state)

Characteristics of a silicon atom

- 1) There are four electrons in the outermost shell of a silicon atom and they are shared between four other neighboring atoms to form four covalent bonds.
- 2) At very low temperature, pure silicon crystal is an insulator (because all bonds in the crystal are unbroken) and has a high resistance to current flow (**no free electrons at 0K**) (**at 0K, Silicon behaves as a perfect insulator**)

**To increase the conductivity of semiconductors, there are two methods**

- 1) Increasing the temperature
- 2) Doping process

First: increasing the temperature

- 1) By increasing the temperature some of the covalent bonds are broken and some of the electrons are freed,
- 2) When the electron is freed, it leaves behind a vacancy in the broken bond (This vacancy is called a **hole**).
- 3) The atom is neutral, and then the absence of **an electron** (negative charge) led to the appearance of a **hole** (positive charge).
- 4) As the temperature increases, the number of free electrons and holes increases, where the number of free electrons equals the number of free holes in a pure semiconductor.
- 5) If the temperature increases above certain limit state of dynamic equilibrium takes place (called **thermal equilibrium**) a small percentage of bonds are broken, so that a fixed number of free electrons and free holes remains constant at every temperature But not the same electrons and same holes remain free. They reshuffle ,but their number stays constant

Thermal equilibrium (dynamic equilibrium)

(The number of bonds broken per second will be equal to the number of bonds mended per second).

Lesson 2

Doping process

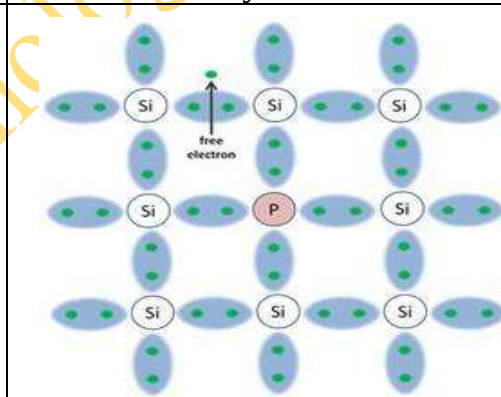
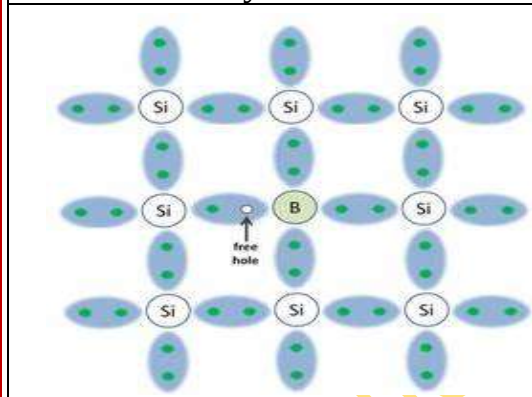
Secondly: Doping Silicon (Germanium)

The doping process

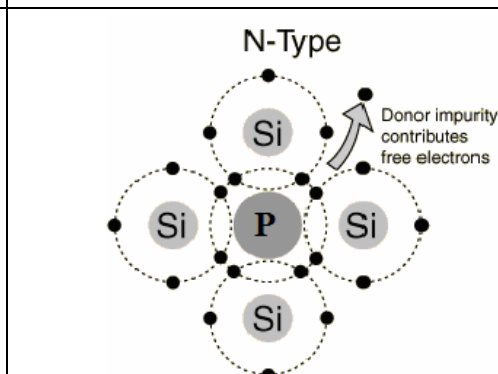
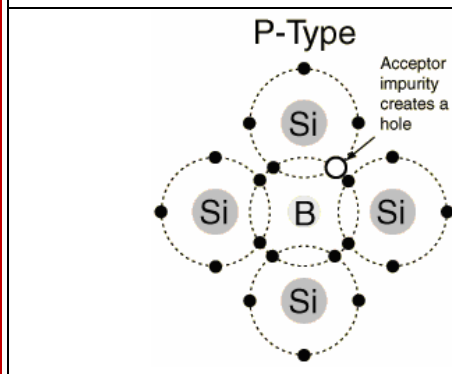
- 1) It is the addition of a small percentage of some suitable impurity atoms in pure semiconductors (**Pentavalent and trivalent impurities are suitable**)
- 2) Increase the electrical conductivity of semiconductors.
- 3) Doping converts intrinsic (pure) semiconductor to extrinsic semiconductor.

Trivalent impurity atoms (acceptor atoms)	Pentavalent impurity atoms (donor atoms)
Atoms having 3 electrons in the outermost shell, (such as Aluminum (Al) or Gallium (Ga) or Boron (B))	Atoms having 5 electrons in the outermost shell (such as Arsenic (As) or Phosphorus (P) or Antimony (Sb))

P-type semiconductor	N-type semiconductor
It is an extrinsic semiconductor which is obtained by doping with trivalent impurity atoms (acceptor atoms) such as Boron (B) atoms in the crystal	It is an extrinsic semiconductor which is obtained by doping with pentavalent impurity atoms (donor atoms) such as Phosphorus (P) atoms in the crystal



The impurity atoms will create a hole which is capable of accepting an electron	The impurity atoms will donate a free electron in the crystal
The holes are majority charge carriers and the free electrons are minority charge carriers	The free electrons are majority charge carriers and the holes are minority charge carriers



Lesson 3

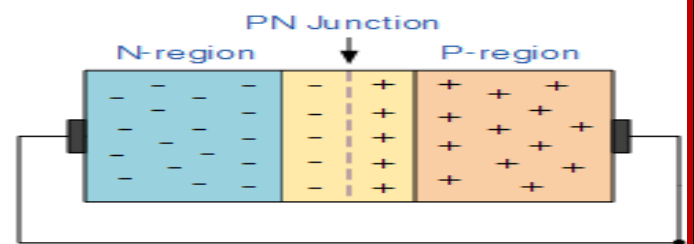
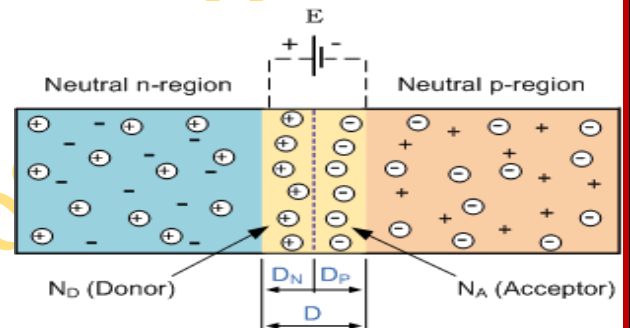
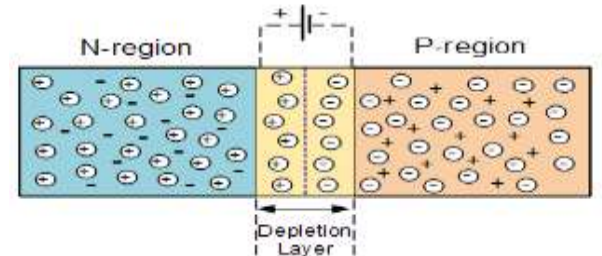
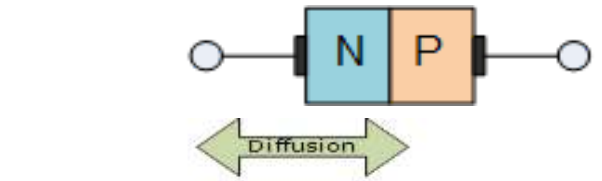
P-N junction

Introduction:

P-N Junction (diode)

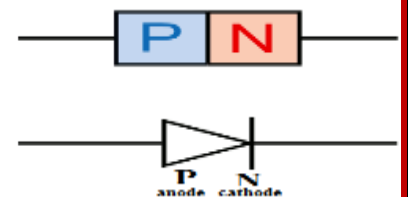
A **PN junction** is formed as shown in Fig. P region has a high concentration of holes and N region contains a large number of electrons. Since **P region** has a high concentration of holes and **N region** contains a large number of electrons,

- a) free electrons and holes cross through the junction by the process of diffusion. During this process, the electrons crossing the junction from N-region into the P region, recombine with holes in the P-region very close to the junction.
- b) Similarly holes crossing the junction from the P-region into the N-region, recombine with electrons in the N-region very close to the junction.
- c) The transfer of some electrons from the N-type region uncovers an equal number of positive donor ions,
- d) and the transfer of some holes from the P-type region uncovers an equal number of negative acceptor ions. This results in a middle region composed of positive ions on one side, and negative ions on the other, while no electrons or holes exist in this region. This region is called **transition (depletion) region**
- e) An electric field is set up, between the donor and acceptor ions in the depletion region directed from the positive ions to the negative ions. This electric field causes a **drift current** to flow in a direction opposite to the **diffusion current**.
- f) At equilibrium, the forward current is balanced with a reverse current, so that the net current is zero



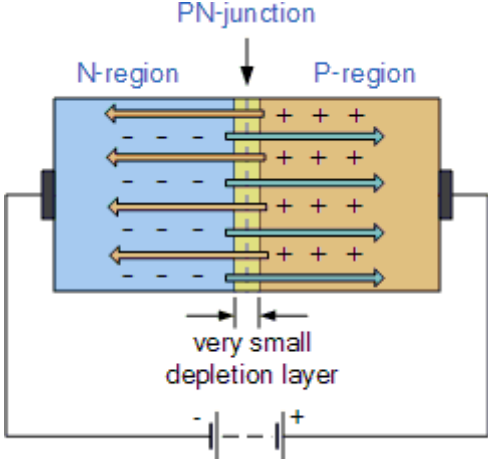
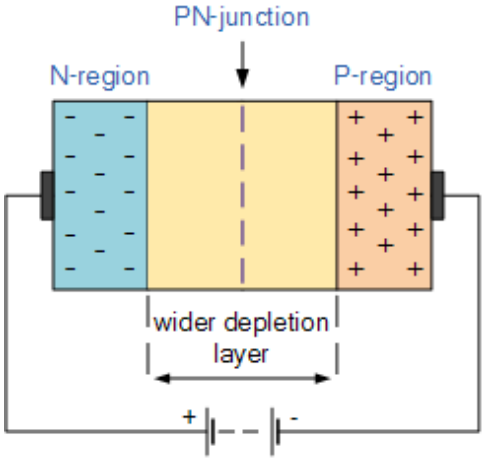
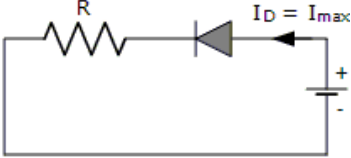
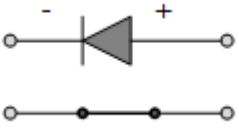
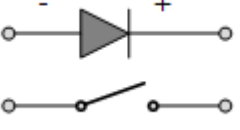
Note that

A diode is represented with the following symbol



Forward Connection (forward bias) and Reverse Connection (Reverse bias)

There are two types of connection

Forward Connection (forward bias)	Reverse Connection (Reverse bias)
The p-type is connected to the positive terminal and the n-type is connected to the negative terminal of a battery.	The p-type is connected to the negative terminal and the n-type is connected to the positive terminal of a battery.
	
	
a) The battery voltage greater than the junction voltage. b) The depletion layer is narrow (because the field due to the battery is opposite to the internal field) c) the resistance of diode decreases Hence a large current flows through the diode	a) The battery voltage is lower than the junction voltage. b) The depletion layer is wide (because the field due to the battery will aid the internal field) c) the resistance of diode increases Hence only a very small current (leakage current) flows through the diode.
The diode in such case work as a closed switch 	The diode in such case work as an open switch 

Notes

- It is a device that allows current to flow in one direction only (unidirectional)
- When a forward voltage is applied, the diode conducts; and when a reverse voltage is applied, there is no conduction.
- It does not obey Ohm's law

Uses of diode**1) Current rectifier (A p-n diode is important in rectification)**

- a) Half-wave rectification
- b) Full-wave rectification

Rectification process

is a process to convert an alternating current into a direct current by using a diode.

2) It is used in charging car batteries, and mobile .batteries, where AC is converted to DC**3) Switch**

The action of the p - n junction is like a switch, which is closed in the forward connection (conducting) and open (non- conducting) in the reverse connection

Very important note

You can use an ohmmeter to distinguish between the ohmic resistance and p-n junction

In case of ohmic resistance

The magnitude of the current is the same, whether or not the voltage polarity is reversed (symmetrical characteristic).

In case of P-N junction**a) In the forward connection**

The diode should have a small resistance hence a large current will flow through the diode

b) In the reverse connection

The diode should have a large resistance hence only a very small current (leakage current) flows through the diode.

Lesson 4

Transistor

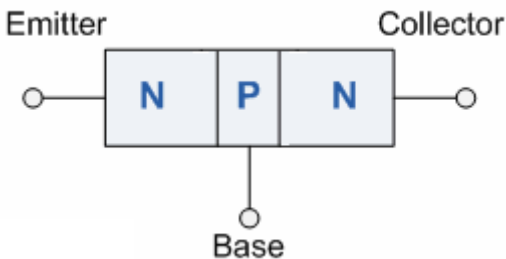
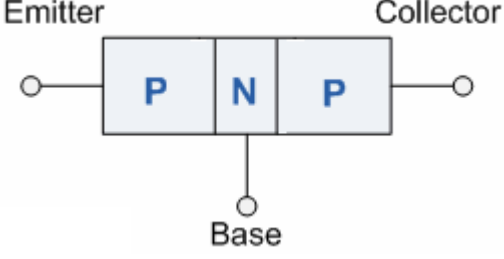
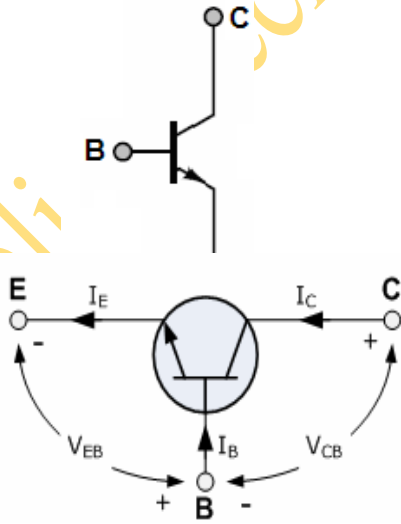
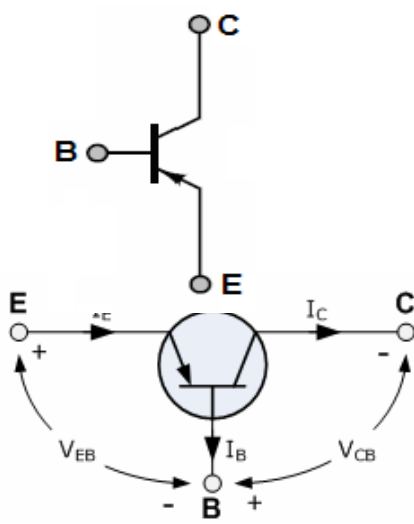
Introduction:

Transistor

- a) The transistor was conceived in 1955 by Bardeen, Shockley and Brattain
- b) There are many types of transistors, but we focus here on bipolar junction transistor (BJT)
- c) A bipolar junction transistor (BJT) consists of three regions of doped semiconductors.

TYPES OF BJT

There are two types of bipolar junction transistors NPN and PNP

Point of comparison	NPN transistor	PNP transistor
	An N- region followed by a P region then N- region	A P- region followed by a N- region then P- region
Construction		
Symbol		

EMITTER

- a) Emitter is a first terminal of bipolar junction transistor
- b) **It is heavily doped** because it emits charge carriers through the thin base layer to be collected by the collector

BASE

- a) Base is a second terminal of bipolar junction transistor
- b) **it is doped lightly** because it is used as a filter or controller in bipolar junction transistor
- c) It contains a small quantity of charge carriers

COLLECTOR

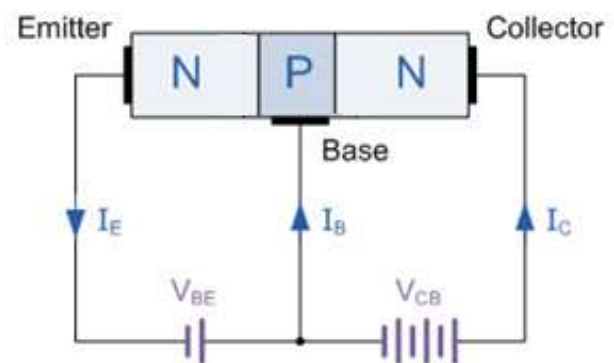
- a) Collector is a third terminal of bipolar junction transistor
- b) **it is doped moderately** because it collects charge carriers which emits by emitter and passes through base.

Note that:

- 1) **Emitter base** is always **forward biased** because these two junction work like a simple diode PN-junction
- 2) **Collector base** is always **reverse biased** because negative terminal pulls all electrons toward itself and collector also carries negative charge
- 3) The action of NPN transistor is similar to that of PNP transistor

NPN transistor**Theory of action**

- 1) Electrons are emitted from the negative emitter (N) diffusing to the base (P), and then they are picked up by the negative collector (N).
- 2) A portion of these electrons are lost by recombination with holes in the base (P) region.
- 3) The base must have narrow width to decrease the chance of recombination.
- 4) If the emitted electron is **I_E (emitter current)** and the portion that reaches the collector region is **I_C (collector current)** then the portion lost in the base by recombination with holes is **I_B (very small base current)**



- 1) Current flowing in the emitter region due to the flow of charges is called emitter current (I_E).
- 2) Current produced for the recombination of electrons and holes is called base current (I_B).
- 3) The current generated in the collector region due to the flow of charges is called collector current (I_C).

The relation between I_E , I_B and I_C :

the emitter current is the sum of collector current and base current (the total emitter current does not flow to the collector region because there is a small current lost in the base due to recombination)(i.e. the collector current is less than the emitter current)
(i.e. $I_E > I_C > I_B$)

$$I_E = I_C + I_B$$

This equation is true regardless of transistor type or transistor configuration.

If

$$I_C = \alpha_e I_E$$

$$I_B = (1 - \alpha_e) I_E$$

Note that:

- 1) α_e is **close to unity** (one) due to the very narrow base region (because only very small portion is lost to recombination)
- 2) α_e represents the percentage of **electrons diffusing across the junction**
- 3) any small change in the base current (I_B), will result in a much larger change in the collector current (I_C). Then, small changes in current flowing in the base will thus control the current in the emitter-collector circuit

Transistor as an amplifier**The basic idea of the transistor action as an amplifier**

If the base current is a small electrical signal, its effect appears amplified in the collector current

any small change in the base current (I_B), will result in a much larger change in the collector current (I_C).
($\Delta I_C \gg \Delta I_B$)

$$I_C = \alpha_e I_E$$

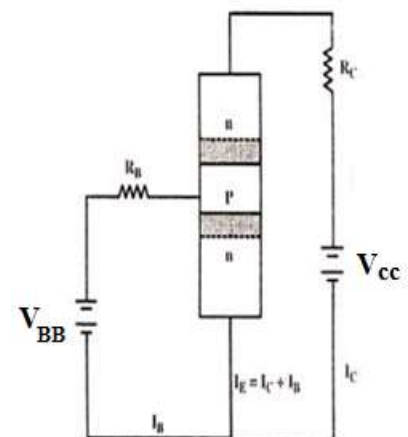
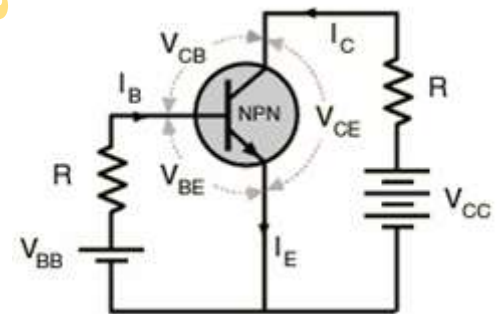
$$I_B = (1 - \alpha_e) I_E$$

$$\beta = \frac{I_C}{I_B} = \frac{\alpha_e I_E}{(1 - \alpha_e) I_E} = \frac{\alpha_e}{1 - \alpha_e}$$

β is called the current gain

The current gain

(It is the ratio between the collector current and the base current)



Lesson 5

Logic gates

Introduction:

Electronic components and devices are the building blocks for all electronic systems. Some of these components are simple, e.g., resistor (R), inductor (L). Some are more complex, such as pn junction (diode), transistor. There are also other specialized devices, such as optoelectronic and control devices. Semiconductors from which most of these devices are made are known to be sensitive to environmental conditions, such as light, heat, pressure, radiation and chemical pollution. That is why they are used as sensors or means for measuring external stimuli. Using these sensors, we can measure the intensity of incident light, temperature, pressure, humidity, pollution, radiation, etc.

Analog electronics

In electronics circuits like amplifiers, the signal (current or voltage) has been in the form of continuous, time-varying voltage or current.

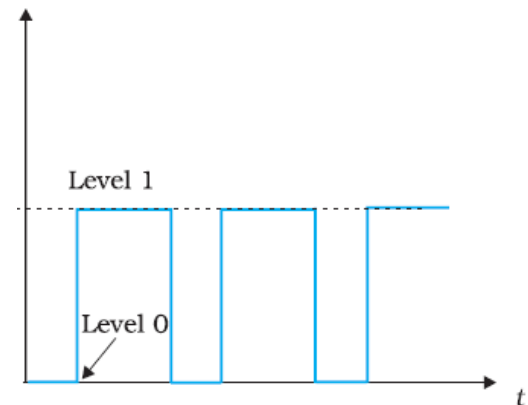
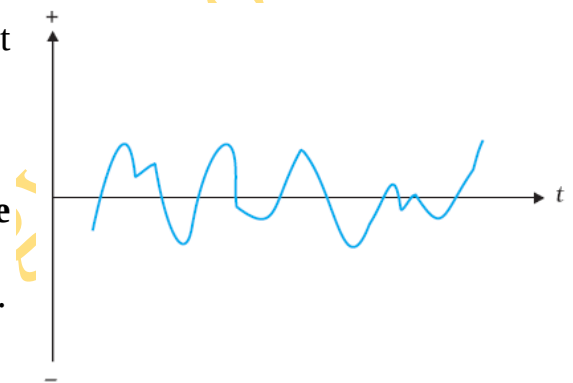
(All electronic systems deal with natural quantities and convert them to continuous electrical signals are called analogue signals).

- 1) A microphone converts sound to an electrical signal.
- 2) A video camera converts an image to electrical signals
- 3) In TV, the image (video) and sound (audio) are transformed into electrical signals

Digital electronics

A new branch of electronics has developed, namely, In this case, the electrical signal is not transmitted continuously (all values are allowed), but is coded, such that the signal is in terms of one of two possible values representing two states 0 or 1 (binary system).

A light switch in your house can be used as an example of a digital circuit. The light is either ON or OFF depending on the switch position. When the light is ON, the output value is '1'. When the light is OFF the output value is '0'. The inputs are the position of the light switch. The switch is placed either in the ON or OFF position to activate the light.

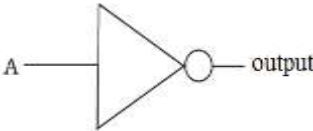


Logic gates

- Systems depend on digital circuits, called logic gates. These are the circuits that perform logic operations.
- Logic gates are used in calculators, digital watches, computers, robots, industrial control systems, and in telecommunications.
- The five common logic gates used are NOT, AND, OR, NAND, NOR. Each logic gate is indicated by a symbol and its function is defined by **a truth table**

NOT gate

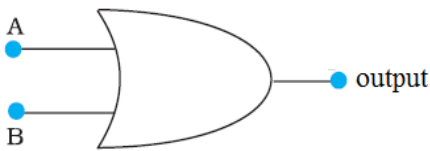
This is the most basic gate, with one input and one output

symbol	Truth table	
	A	Output
	0	1
	1	0

OR Gate

An OR gate has two or more inputs with one output

The electronic circuit that gives a high output (1) if **one or more** of its inputs are high

symbol	Truth table		
	A	B	Output
	0	0	0
	0	1	1
	1	0	1
	1	1	1

AND Gate

An AND gate has two or more inputs with one output

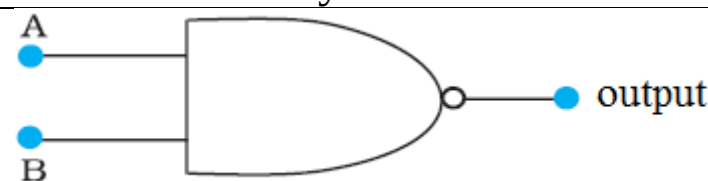
The electronic circuit that gives a high output (1) only if **all** its inputs are high

symbol	Truth table		
	A	B	Output
	0	0	0
	0	1	0
	1	0	0
	1	1	1

NAND Gate

It is an AND gate followed by NOT gate, has two or more inputs with one output

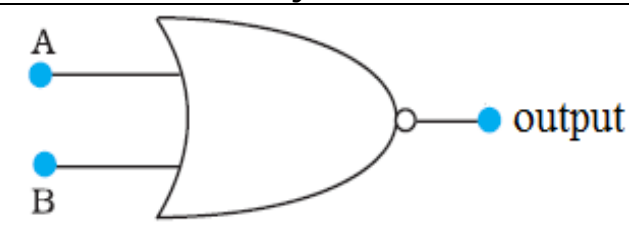
The electronic circuit that gives a low output (0) only if **all** its inputs are high

symbol	Truth table		
	A	B	Output
	0	0	1
	0	1	1
	1	0	1
	1	1	0

NOR Gate

It is an OR gate followed by NOT gate, has two or more inputs with one output

The electronic circuit that gives a high output (1) only if **all** its inputs are low

symbol	Truth table		
	A	B	Output
	0	0	1
	0	1	0
	1	0	0
	1	1	0